Counting subsets of bounded separation complexity

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Editorial note. Faithful transcription from official CCC 2024 CC BY 4.0 PDF; original notation, indexing and dimensions retained. Author TeX was not located. No author proof is replaced or extended; standard cited prior results are retained as prerequisites, and bibliography entries keep their original numbers.

Allowed standard prerequisites (editorial).

- Single existential-block quantifier elimination bound. Basu, Pollack and Roy [1], Theorem 14.16, exact source statement (4) retained

6.1 Extended formulations and separation complexity (author source p. 13:8)

Let us first make a short detour into extended formulations of convex polyhedra. A polyhedron $P \subseteq \mathbb{R}^r$ is a (possibly unbounded) set defined by a finite number of linear constraints. Following [20,18,2], define the extension complexity of P, $\mathrm{xc}(P)$, as the smallest s such that P is a linear projection of a polyhedron $Q \subseteq \mathbb{R}^m$ where Q can be defined using s inequalities (and any number of equalities). Observe that P with extension complexity s can be expressed in the standard form

$$x \in P \text{ iff } \exists y \in \mathbb{R}^s \; Cx + Dy = b, \; y \geq 0,$$

where $C \in \mathbb{R}^{t \times r}$, $D \in \mathbb{R}^{t \times s}$ and $b \in \mathbb{R}^t$ for some t. Let V be a finite subset of $\mathbb{R}^r$. Given $A \subseteq V$, its separation complexity, $\mathrm{sep}_V(A)$, is the minimum $\mathrm{xc}(P)$ over all polyhedra $P \subseteq \mathbb{R}^r$ with[1]

$$P \cap V = A;$$

such a P is called a separating polyhedron for A. In other words, $\mathrm{sep}_V(A)$ is the smallest s so that we can distinguish points in A from points in $V \setminus A$ by means of a linear program with s inequalities. Moreover, such a program can be rewritten as

$$x \in A \text{ iff } (x \in V \text{ and } \exists y \in \mathbb{R}^s \; Cx + Dy = b, \; y \geq 0).$$

[1] If no such polyhedron exists, which may happen if V is not convexly independent, we set $\mathrm{sep}_V(A) := \infty$.

The notion of separation complexity has been studied in [7,8,9] in the case when $V = \{0,1\}^n$ is the Boolean cube. The following theorem is of independent interest and can be seen as an extension of similar results in [7,9]. The proof is a considerable simplification of the previous ones.

Theorem 13 (author source p. 13:8). *Let V be a non-empty finite subset of $\mathbb{R}^r$. Given a parameter $s \geq 1$, let $\mathcal{A}$ be the collection of subsets A of V with $\mathrm{sep}_V(A) \leq s$. Then*

$$|\mathcal{A}| \leq 2^{O(s(r+s)^2 \log |V|)}.$$

A Proof of Theorem 13 (author source pp. 13:11–13:12)

The proof uses known results on quantifier elimination which we first outline. We follow the monograph of Basu, Pollack and Roy [1]. Theorem 13 requires an elimination of only a single block of existential quantifiers, so we focus on this case only. For $b \in \mathbb{R}$, let

$$\mathrm{sgn}(b) := \begin{cases} 1, & b > 0, \\ 0, & b = 0, \\ -1, & b < 0. \end{cases}$$

Given $b = \langle b_1, \ldots, b_m \rangle \in \mathbb{R}^m$, let $\mathrm{sgn}(b) := \langle \mathrm{sgn}(b_1), \ldots, \mathrm{sgn}(b_m) \rangle \in \{-1,0,1\}^m$. Let $F = F(z,y)$ be a sequence of m polynomials $f_1, \ldots, f_m \in \mathbb{R}[z,y]$ in variables $z = \{z_1, \ldots, z_{k_1}\}$ and $y = \{y_1, \ldots, y_{k_2}\}$. Given $a \in \mathbb{R}^{k_1}$, define $\mathrm{SGN}_1(F,a) \subseteq \{-1,0,1\}^m$

$$\mathrm{SGN}_1(F,a) := \{\mathrm{sgn}(F(a,b)) : b \in \mathbb{R}^{k_2}\}.$$

Let

$$\mathrm{SGN}(F) := \{\mathrm{SGN}_1(F,a) : a \in \mathbb{R}^{k_1}\}.$$

Theorem 14.16 from [1] provides the following bound on the size of SGN:

Theorem ([1]). *If every polynomial in F has degree at most d then*

$$|\,\mathrm{SGN}(F)| \leq m^{(k_1+1)(k_2+1)} d^{O(k_1)O(k_2)}. \tag{4}$$

We now apply this result to the case of Theorem 13. Let $V, s, \mathcal{A}$ be as in the assumption. Every $A \in \mathcal{A}$ can be described by a linear system with s inequalities. Namely, for every $x \in V$,

$$x \in A \text{ iff } \exists y \in \mathbb{R}^s \; Cx + Dy = b, \; y \geq 0, \tag{5}$$

where $C \in \mathbb{R}^{t \times r}$, $D \in \mathbb{R}^{t \times s}$ and $b \in \mathbb{R}^t$. Since $Cx + Dy = b$ is a system of equations in $r+s$ variables x, y, we can also assume $t = r + s$. Let us view the parameters C, D, b in (5) as variables. Let z be the set of these variables, of size $k_1 = (r+s)(r+s+1)$. Given $v \in V$, let $F_v(z,y)$ be the sequence of $(r+s)$ polynomials

$$Cv + Dy - b$$

in variables z and $y = \{y_1, \ldots, y_s\}$. Let $F(z, y)$ be the union of $F_v(z, y)$ over all $v \in V$, together with the polynomials $y_1, \ldots, y_s$. Hence F consists of $m = s + |V|(r + s)$ polynomials of degree at most two. $F(z, y)$ is set up so that

$$|\mathcal{A}| \leq |\operatorname{SGN}(F)|.$$

To see this, observe that whenever the parameters z are fixed, the set $A \subseteq V$ given by (5) is uniquely determined by $\operatorname{SGN}_1(F(z, y))$. Since every $A \in \mathcal{A}$ is obtained by some fixing of the parameters, we indeed obtain $|\mathcal{A}| \leq |\operatorname{SGN}(F(z, y))|$. Finally, we can apply (4) to estimate $|\operatorname{SGN}(F)|$ with $m = s + |V|(r + s)$, $k_1 = (r + s)(r + s + 1)$, $k_2 = s$, and $d = 2$. To simplify the expression, we can assume $s + r \leq |V|$; otherwise the upper bound asserted in Theorem 13 exceeds the trivial bound $|\mathcal{A}| \leq 2^{|V|}$. This means that $m \leq 2|V|^2$. If we loosen the bound (4) as $|\operatorname{SGN}(F)| \leq (dm)^{O(k_1)O(k_2)}$, we obtain (recall that $s \geq 1$)

$$|\operatorname{SGN}(F)| \leq 2^{O(s(s+r)^2 \log |V|)},$$

as required.

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